WORKED SOLUTION

STEP II 2015, Question 2

Pure • Euclidean geometry

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Read the question before you go on. It is in the STEP Support Programme's question database:

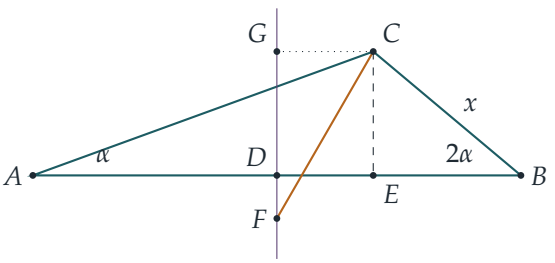
step.maths.org/questions/15-s2-q2

The key idea

The first result is the sine rule plus the triple-angle formula for $\sin 3\alpha$. For DE, measure everything along AB from one end: the right-angled triangles on either side of CE give AE and BE directly. The answer for DE turns out to be remarkably simple, and that simplicity is exactly what makes the trisection work: it tells us how far C is from the perpendicular bisector, and comparing that with $CF = x$ pins down an angle.

Solution

Throughout, the angle at C is $\angle ACB = \pi - 3\alpha$, and since 2α is acute we have $0 < \alpha < \frac{\pi}{4}$.



The length AB

By the sine rule, using $\sin(\pi - 3\alpha) = \sin 3\alpha$,

$$\frac{AB}{\sin 3\alpha} = \frac{x}{\sin \alpha}.$$

The triple-angle formula $\sin 3\alpha = 3\sin\alpha - 4\sin^3\alpha$ then gives, dividing by $\sin\alpha \neq 0$,

$$AB = x\,\frac{3\sin\alpha - 4\sin^3\alpha}{\sin\alpha} = (3 - 4\sin^2\alpha)\,x,$$

as required.

The length DE

Both angles at A and B are acute, so the foot E of the perpendicular from C lies between A and B. In the right-angled triangle CEB,

$$BE = x\cos 2\alpha, \qquad CE = x\sin 2\alpha.$$

Hence

$$AE = AB - BE = x(3 - 4\sin^2\alpha - (1 - 2\sin^2\alpha)) = x(2 - 2\sin^2\alpha) = 2x\cos^2\alpha,$$
$$AD = \tfrac{1}{2}AB = x(\tfrac{3}{2} - 2\sin^2\alpha).$$

Subtracting,

$$AE - AD = x(2 - 2\sin^2\alpha - \tfrac{3}{2} + 2\sin^2\alpha) = \tfrac{1}{2}x.$$

This is positive, so E lies between D and B, and

$$DE = \tfrac{1}{2}x,$$

independent of α.

The trisection

Let G be the foot of the perpendicular from C to the perpendicular bisector DF of AB. Since CE and GD are both perpendicular to AB, $CGDE$ is a rectangle, so $CG = DE = \frac{1}{2}x$ and CG is parallel to AB, i.e. perpendicular to CE.

Ask yourself How do we turn $CF = x$ into an angle?

The triangle CGF has a right angle at G, hypotenuse $CF = x$ and side $CG = \frac{1}{2}x$. So

$$\cos\angle GCF = \frac{CG}{CF} = \frac{1}{2}, \qquad \text{so} \qquad \angle GCF = \frac{\pi}{3}.$$

There are two points of the bisector at distance x from C, one either side of G. Measured from the ray CE (pointing down towards AB) round towards A, the ray CG is at angle $\frac{\pi}{2}$, so the two candidate rays CF are at angles $\frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$ (the point below G) and $\frac{\pi}{2} + \frac{\pi}{3} = \frac{5\pi}{6}$ (the point above G).

Now decide which one is on the same side of AC as B. In the right-angled triangle ACE, $\angle ACE = \frac{\pi}{2} - \alpha$, so the ray CA is at angle $\frac{\pi}{2} - \alpha$ from CE on this same side. The rays from C on B's side of the line AC are those that, measured this way, are at less than $\frac{\pi}{2} - \alpha$ (or on the other side of CE, towards B). Because $\alpha < \frac{\pi}{4}$,

$$\frac{\pi}{6} < \frac{\pi}{4} < \frac{\pi}{2} - \alpha < \frac{5\pi}{6},$$

so the point below G is on B's side and the point above G is not. Thus F is the lower point, $\angle ECF = \frac{\pi}{6}$, and the ray CF lies inside the angle ACE, hence inside $\angle ACB$. Therefore

$$\angle ACF = \angle ACE - \angle ECF = \left(\frac{\pi}{2} - \alpha\right) - \frac{\pi}{6} = \frac{\pi}{3} - \alpha = \frac{1}{3}(\pi - 3\alpha) = \frac{1}{3}\angle ACB.$$

So CF cuts off one third of $\angle ACB$ next to CA (and $\angle FCB = \frac{2}{3}\angle ACB$): the line FC trisects the angle ACB.

Checks

- With $\alpha = \frac{\pi}{6}$ the triangle has angles 30°, 60°, 90°, and $AB = (3-1)x = 2x$, the familiar hypotenuse. Then $AE = 2x\cos^2 30^\circ = \frac{3}{2}x$ and $AD = x$, so $DE = \frac{1}{2}x$. The angle at C is 90°, and $\angle ACF = 60^\circ - 30^\circ = 30^\circ$, a third of it.
- With $\alpha = 20^\circ$ and $x = 1$ we get $AB \approx 2.532$ and $\angle ACB = 120^\circ$; constructing F directly from coordinates gives $\angle ACF = 40^\circ$. All three results were also confirmed by computer for thousands of random triangles.

Common slips

- Taking F above G. That point is also distance x from C, but it lies on the far side of AC, which is why the question specifies the side.
- Forgetting to check that E lies between A and B (and on which side of D) before writing DE as a difference of lengths.

Beyond STEP

Trisecting a general angle with ruler and compasses is impossible, a fact proved with field theory: trisecting 60° would require constructing a root of an irreducible cubic, and constructible lengths only ever satisfy towers of quadratics. Here the triangle supplies extra information (the angle at B is twice that at A), and that is what lets a simple construction do the job. For more elegant triangle geometry see Coxeter and Greitzer, *Geometry Revisited*; for the impossibility proof, Allenby, *Rings, Fields and Groups*.

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